$\underline{V} = \underline{\nabla} \Phi$

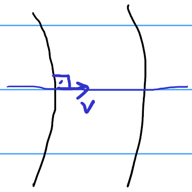
$\underline{V} = \begin{bmatrix} u \\ v \end{bmatrix}$

Función Potencial

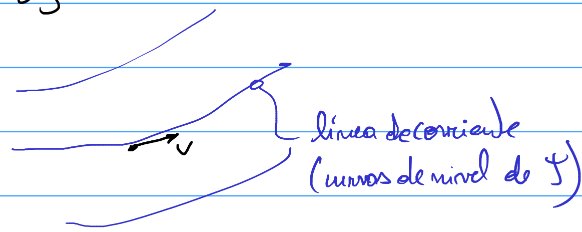
$u = \frac{\partial \Phi}{\partial x}, \quad v = \frac{\partial \Phi}{\partial y},$

Función línea de corriente

$u = \frac{\partial \psi}{\partial y} \qquad v = -\frac{\partial \psi}{\partial x}$



Curvas de nivel Φ

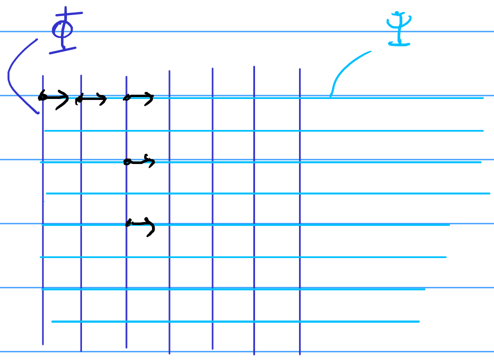


b)

$\Phi = x \qquad \Psi = y$

$u = \frac{\partial \Phi}{\partial x} = 1 \qquad u = \frac{\partial \psi}{\partial y} = 1$

$v = \frac{\partial \Phi}{\partial y} = 0 \qquad v = -\frac{\partial \psi}{\partial x} = 0$



Vorticidad

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial v_j}{\partial x_i} - \frac{\partial v_i}{\partial x_j}\right)$$

En 2D:

$$\underline{\underline{\Omega}} = \begin{bmatrix} 0 & \Omega_{12} \\ -\Omega_{12} & 0 \end{bmatrix}$$

Funcion Potencial

$$\Omega_{12} = \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2}\right) = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = \frac{1}{2}\left[\frac{\partial}{\partial x}\cancel{\left(\frac{\partial \phi}{\partial y}\right)} - \frac{\partial}{\partial y}\cancel{\left(\frac{\partial \Phi}{\partial x}\right)}\right] = 0$$

Función de Corriente

$$\Omega_{12} = \frac{1}{2}\left(\frac{\partial}{\partial x}\left(\frac{-\partial \psi}{\partial x}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial y}\right)\right) = -\frac{1}{2}\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right)$$

tasa de deformaciones

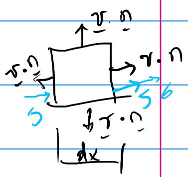
$$\underline{\underline{V}} = \begin{bmatrix} V_{11} & V_{12} \\ V_{12} & V_{22} \end{bmatrix}$$

$$V_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)$$

$V_{11} = \frac{\partial u}{\partial x} \qquad V_{22} = \frac{\partial v}{\partial y}$

$$V_{12} = \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)$$

$$V_{11} = \frac{\partial}{\partial x}(1) = 0 \qquad V_{22} = \frac{\partial}{\partial y}(0) = 0 \qquad V_{12} = \frac{1}{2}\left(\frac{\partial (1)}{\partial y} + \frac{\partial (0)}{\partial x}\right) = 0$$

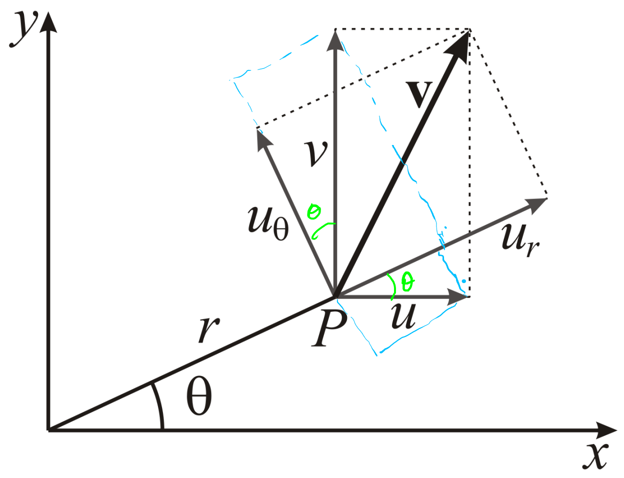


$$\underline{\nabla} \cdot \underline{v} = 0$$

2D

$$\underline{\nabla} \cdot \underline{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial y}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial \psi}{\partial x}\right) = 0$$

$$\underline{\nabla} \cdot \underline{v} = \frac{\partial}{\partial x}\left(\frac{\partial \Phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial \Phi}{\partial y}\right) = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2}$$



$u_\theta = -\sin\theta \cdot u + v \cdot \cos\theta$

$u_r = \cos\theta \cdot u + v \cdot \sin\theta$

$x = r \cdot \cos\theta$

$y = r \cdot \sin\theta$

$$\Phi(r,\theta) = \Phi^*\big(x(r,\theta), y(r,\theta)\big)$$

$$\frac{\partial \Phi(r,\theta)}{\partial r} = \underbrace{\frac{\partial \Phi^*}{\partial x}(x,y)}_{u} \cdot \overbrace{\frac{\partial x}{\partial r}}^{\frac{\partial (r \cdot \cos\theta)}{\partial r}} + \frac{\partial \Phi^*}{\partial y}(x,y) \cdot \frac{\partial y}{\partial r} =$$

$$= u \cdot \cos\theta + v \cdot \sin\theta = \boxed{u_r} \Rightarrow u_r = \frac{\partial \Phi(r,\theta)}{\partial r}$$

$$\frac{\partial \Phi(r,\theta)}{\partial \theta} = \frac{\partial \Phi^*(x,y)}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial \Phi^*(x,y)}{\partial y} \cdot \frac{\partial y}{\partial \theta} =$$

$$= u \cdot -r \cdot \sin\theta + v \cdot r \cdot \cos\theta =$$

$$= r(-u \cdot \sin\theta + v \cdot \cos\theta) = r \cdot u_\theta \Rightarrow u_\theta = \frac{1}{r} \frac{\partial \Phi(r,\theta)}{\partial \theta}$$